

## Coding & Solution

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 1 <sup>st</sup> july            |
| <b>Team ID</b>       | xxxxxxx                         |
| <b>Project Name</b>  | Credit card approval prediction |
| <b>Maximum Marks</b> | 5 Marks                         |

### Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

#### Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

### Solution Summary

| Field                           | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Repository Link / URL</b>    | <a href="https://github.com/swathipraneethanukala/credit-card-approval-prediction">https://github.com/swathipraneethanukala/credit-card-approval-prediction</a>                                                                                                                                                                                                                                                                                                       |
| <b>Programming Language(s)</b>  | Python 3.10+, HTML5, CSS3                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Framework(s) Used</b>        | Flask (v3.0+)                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Key Features Implemented</b> | <input type="checkbox"/> <b>Categorical Feature Imputation &amp; Cleaning</b><br><input type="checkbox"/> <b>Target Variable Binarization Mapping</b><br><input type="checkbox"/> <b>Minority-Class Weight Imbalance Correction</b><br><input type="checkbox"/> <b>Automated Feature Encoding Pipeline</b><br><input type="checkbox"/> <b>Real-time Local Web Inference Dashboard</b><br><input type="checkbox"/> <b>Robust Error Handling &amp; Input Validation</b> |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pending / Incomplete Features</b> | <ul style="list-style-type: none"> <li>* Cloud production hosting (currently running entirely on a localized developer workspace sandbox).</li> <li>* Feature explainability charts (such as rendering visual SHAP or LIME force plots directly on the web result screen).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Setup / Run Instructions</b>      | <p><b>1. Clone the Project Repository:</b><br/> <a href="https://github.com/swathipraneethanukala/credit-card-approval-prediction">https://github.com/swathipraneethanukala/credit-card-approval-prediction</a></p> <p><b>2. Install Required Dependencies:</b><br/>         pip install -r requirements.txt</p> <p><b>3. Execute Model Training &amp; Binary Serialization Pipeline:</b><br/>         python train_model.py<br/> <i>(This reads the raw CSV files, trains the Random Forest model, and generates the model.pkl binary asset).</i></p> <p><b>4. Launch the Local Web Application Server:</b><br/>         python app.py</p> <p><b>5. Access UI Dashboard:</b><br/>         Open your browser and navigate to the local WSGI server port at:<br/> <a href="http://127.0.0.1:5000/">[http://127.0.0.1:5000/](http://127.0.0.1:5000/)</a></p> |

### Code Quality Checklist

| S.No | Criteria                                               | Status (Yes / No) |
|------|--------------------------------------------------------|-------------------|
| 1    | Code is modular and organized into functions / classes | YES               |
| 2    | Meaningful variable and function names are used        | YES               |
| 3    | Code includes comments / documentation where necessary | YES               |
| 4    | Error handling is implemented for critical operations  | YES               |
| 5    | The application runs without critical errors           | YES               |
| 6    | Code is committed to a version control repository      | YES               |

### **Additional Notes / Comments:**

- **Balanced Minority Recall:** The primary engineering highlight of this repository is the configuration of class weights within the ensemble classifier, which ensures that high-risk candidate patterns are detected reliably despite making up a tiny percentage of the raw source records.
- **Synchronous Inference Speed:** By loading the serialized model.pkl binary pipeline directly into memory at startup, the local Flask router runs feature transformation arrays and generates prediction strings in under 1.5 seconds per user query.